

6 - 2

- 問1 トリオlein酸グリセリド ( $890 - 2 \times 1 \times 3 = 884$ ) : 10%  
 トリリノール酸 " ( $890 - 2 \times 2 \times 3 = 878$ ) : 80%  
 トリリノレン酸 " ( $890 - 2 \times 3 \times 3 = 872$ ) : 10%

ステアリン酸 2H C=C数  
 3分子からなる  
 油脂

$$\bar{M} = \frac{884 \times \frac{10}{100}}{= 872 + 12} + \frac{878 \times \frac{80}{100}}{= 872 + 6} + \frac{872 \times \frac{10}{100}}{}$$

$$= 872 + 12 \times \frac{10}{100} + 6 \times \frac{80}{100} = 878 = 890 - \frac{12}{= 2H \times 6} \quad C=C数$$

<別解>

油脂 1分子中の脂肪酸3つ中に含まれる不飽和脂肪酸は、

$$\text{olein酸} : 3 \times 0.1 = 0.3 \rightarrow \frac{\times 1}{C=C \times 0.3}$$

$$\text{lin酸} : 3 \times 0.8 = 2.4 \rightarrow \frac{\times 2}{C=C \times 4.8}$$

$$\text{lin酸} : 3 \times 0.1 = 0.3 \rightarrow \frac{\times 3}{C=C \times 0.9} \quad (+ \\ C=C \times 6)$$

$$890 - 2 \times 6 = 878$$

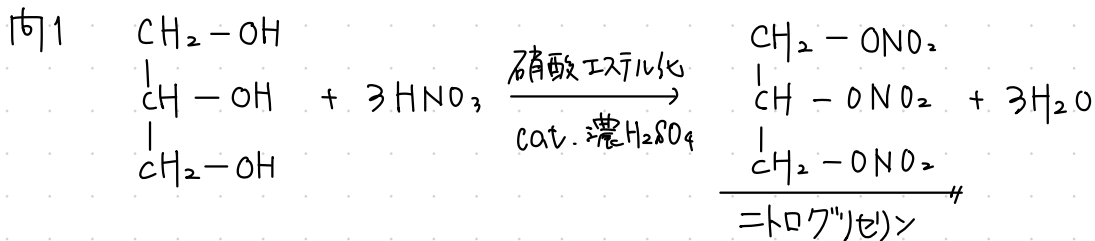
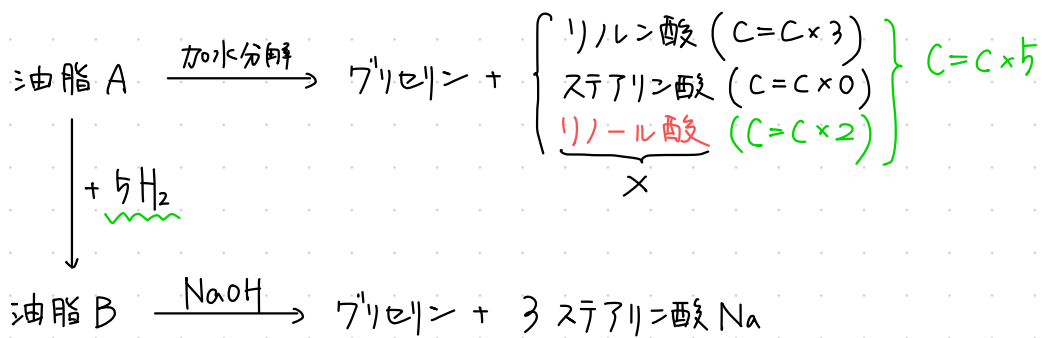
問2

$$\frac{1 \text{ g}}{878 \text{ g/mol}} \times 3 = \frac{4 \text{ L化油} \times 10^{-3} \text{ g}}{56 \text{ g/mol}} \quad \therefore 4 \text{ L化油} = 191$$

問3. この油脂は、1分子あたり平均6つのC=Cをもつから、

$$\frac{100 \text{ g}}{878 \text{ g/mol}} \times 6 \times 22.4 \text{ L/mol} = 1.53 \times 10 \text{ L}$$

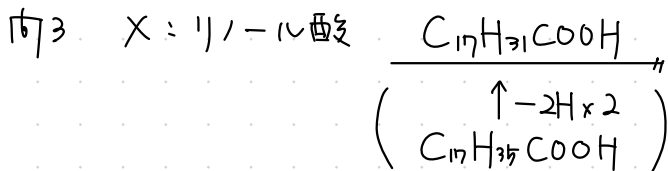
6- [3]



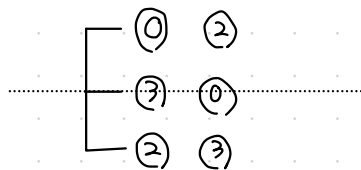
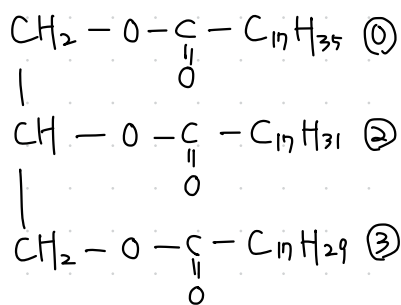
問2  $\left( \text{油脂 A} = 890 - 2\text{H} \times 5 = 880 \right) \Rightarrow \text{油脂 B} : 890$

油脂 A : NaOH = 1 : 3 対

$$\frac{100\text{ g}}{890\text{ g/mol}} \times 3 = \frac{W\text{ g}}{40\text{ g/mol}} \quad \therefore W = 13.5\text{ g}$$



問4.

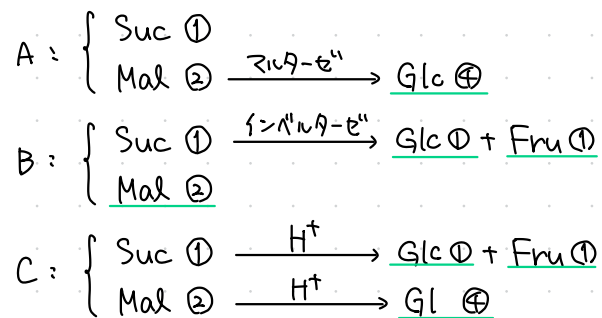


6-4

- 問1 スクロース : (3), A  
 ゼルトース : (1), D  
 セロビオース : (4), B  
 ラクトース : (5), E  
 トレハロース : (2), C

問2 ゼ, セ, ラ

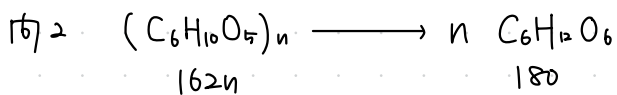
問3



$$4 : 4 : 6 = 2 : 2 : 3$$

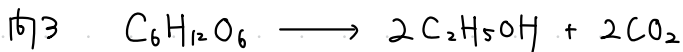
6-5

問1. ① 1 ② 4 ③ 1 ④ 6



$$\frac{162 \text{ g}}{162n \text{ g/mol}} \times n \times 180 \text{ g/mol} = 180 \text{ g}$$

蔗糖 mol                      グルコース mol



$$\frac{180 \text{ g}}{180 \text{ g/mol}} \times 2 \times 46 \text{ g/mol} = 92 \text{ g}$$

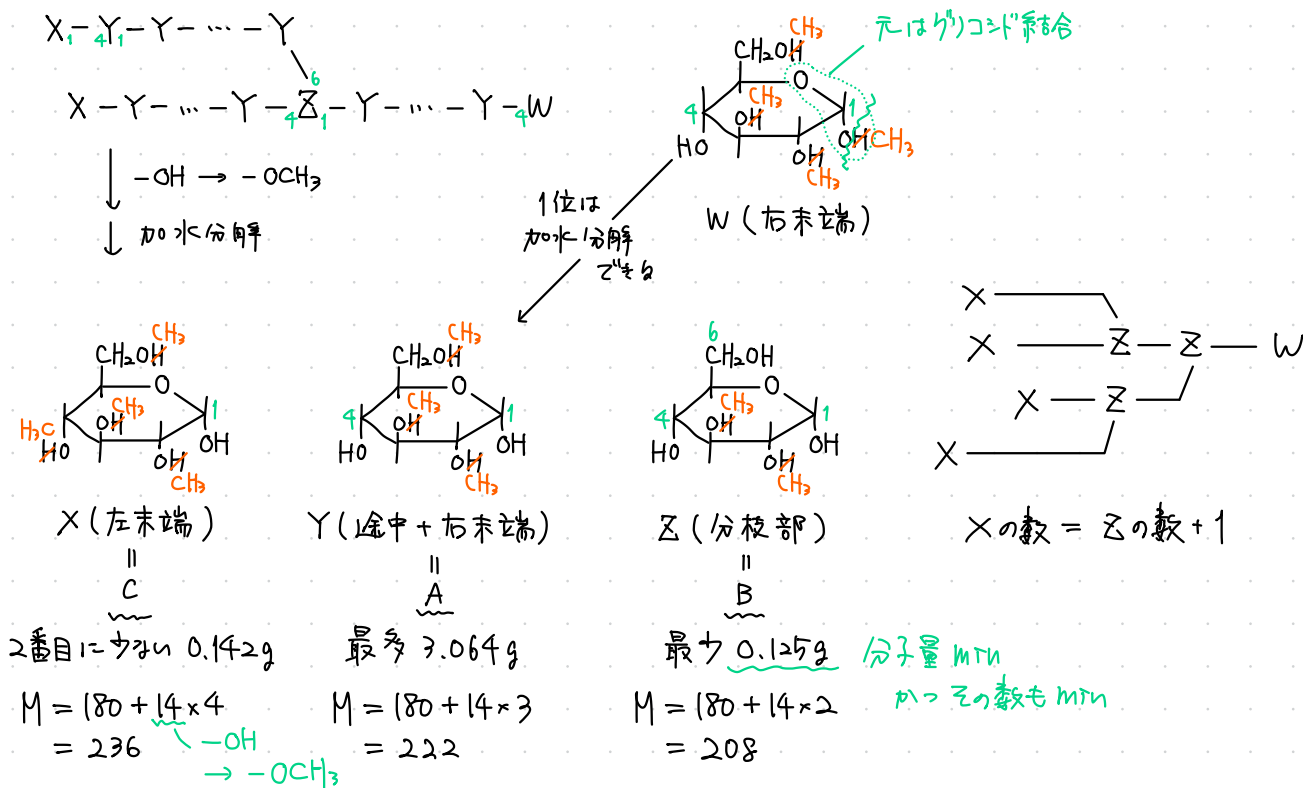
グルコース mol                      C<sub>2</sub>H<sub>5</sub>OH mol

問4. A : グルコース    B : ゼルトース    C : フルクトース    D : スクロース  
 E : ラクトース    F : ガラクトース    G : テンブレン    H : セロース

6-6

①  $162n = 4.05 \times 10^5 \quad \therefore n = \underline{2500} (= 2.5 \times 10^3)$

② ~ ④



$$A : B : C = \frac{3.064g}{222 g/mol} : \frac{0.125g}{208 g/mol} : \frac{0.142g}{236 g/mol}$$

$$\approx 1.38 \times 10^{-2} : \underline{6.00 \times 10^{-4}} : \underline{6.01 \times 10^{-4}}$$

$$\approx \underline{23} : 1 : 1$$

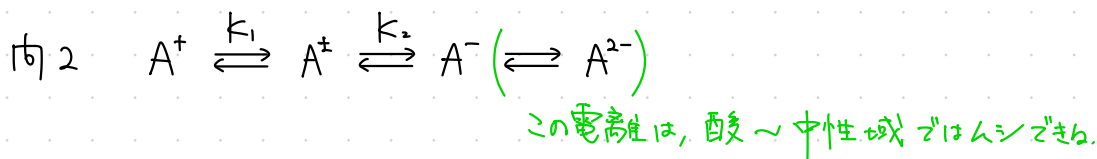
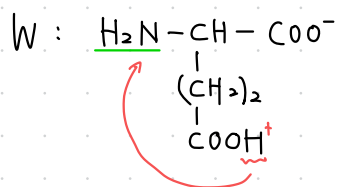
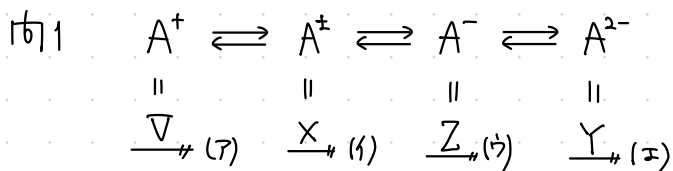
↑ ↑ ほぼ同じにする

→ 2500 中 1 コの Glc 分枝部

$$\text{分枝部} = 2500 \times \frac{1}{25} = \underline{100} (= 1.0 \times 10^2)$$



6 - 10



等電点では,  $[A^+] \times 1 = [A^-] \times 1 + [A^{2-}] \times 2$

$$([A^+] \times 1 + [A^-] \times (-1) + [A^{2-}] \times (-2) = 0)$$

今回は,  $[A^{2-}]$  はムシズキのぞ,  $\underline{[A^+] = [A^-]}$  中性アミノ酸と同じ

$$[H^+] = \sqrt{K_1 K_2} = 10^{-3.3} \quad \underline{\text{pH } 3.3}$$

問3. それぞれの濃度を  $[A^\pm]$  で表す.

$$\begin{aligned}
 [A^+] : [A^\pm] : [A^-] &= \frac{[H^+]}{K_1} \cancel{[A^\pm]} : \cancel{[A^\pm]} : \frac{K_2}{[H^+]} \cancel{[A^\pm]} \\
 &= 10^{-1} : 1 : 10^{-1} \\
 &= 1 : 10 : 1
 \end{aligned}$$

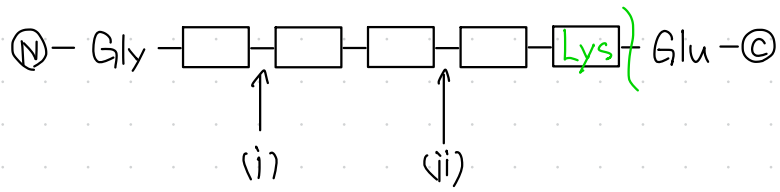
$$\therefore \frac{[A^\pm]}{C} = \frac{10}{1+10+1} = 0.833 \dots \quad \underline{83\%}$$

問4 ①より

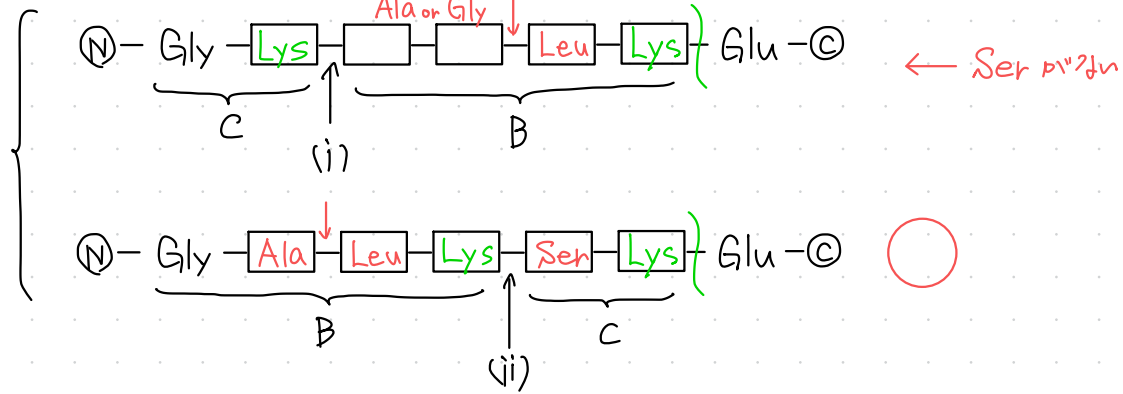
$$\begin{aligned}
 [A^\pm] &= \frac{5}{6} \times 1 \times 10^{-3} \\
 &\doteq \underline{8.3 \times 10^{-4}}
 \end{aligned}$$



6 - 12



↓ -Lys }



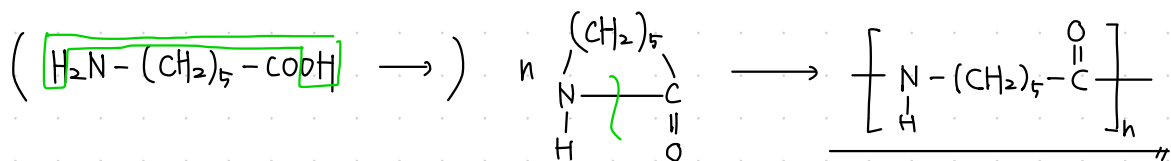
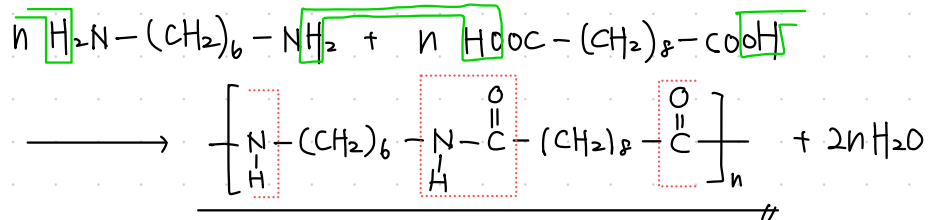




7-1

- 問1 ア. ポリアミド イ. ヘキサメチレンジアミン ウ. 縮合重合  
エ. ε-カプロラクタム オ. 開環重合

問2.

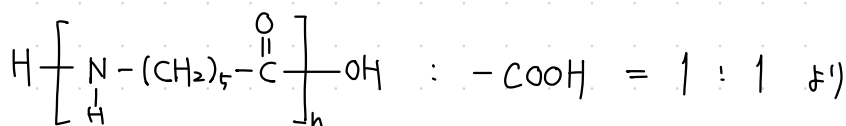


問3  $\Pi V = \frac{w}{M} RT$  より

$$M = \frac{wRT}{\Pi V} = \frac{8.3 \times 10^4}{\quad} = 282n \quad \therefore n = 2.94 \times 10^2$$

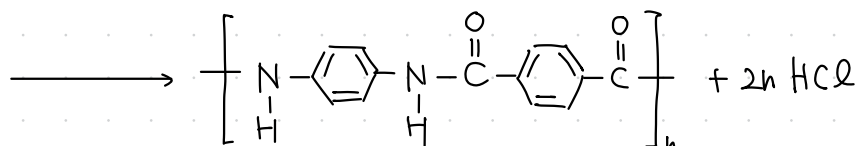
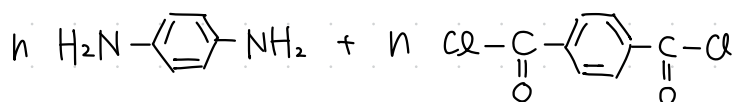
$$\therefore 2n \doteq 5.9 \times 10^2 \quad \#$$

問4



$$\frac{100 \text{ g}}{113n + 18 \text{ g/mol}} = 0.03 \text{ mol} \quad \therefore n \doteq 294 \doteq 2.9 \times 10^2 \quad \#$$

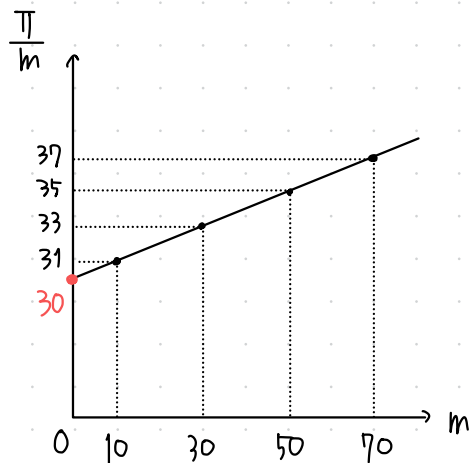
問5 アラミド



17-2

16)2

| $m \text{ [g/L]}$  | (0)                  | 10  | 30  | 50   | 70   |
|--------------------|----------------------|-----|-----|------|------|
| $\Pi \text{ [Pa]}$ |                      | 310 | 990 | 1750 | 2590 |
| $\frac{\Pi}{m}$    | <u>30</u><br>この値を求めた | 31  | 33  | 35   | 37   |

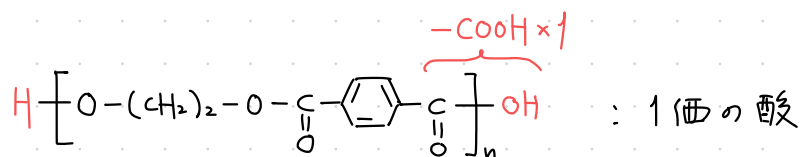


$$\Pi V = \frac{w}{M} RT \quad \therefore \Pi = \underbrace{\frac{w}{V}}_m \cdot \frac{RT}{M}$$

$$\therefore M = \frac{RT}{\Pi/m} \xrightarrow{m \rightarrow 0} \frac{RT}{(\Pi/m)_0} = \frac{8.3 \times 10^3 \times 289}{30} = 7.99 \times 10^4$$

$$= \underline{8.0 \times 10^4}$$

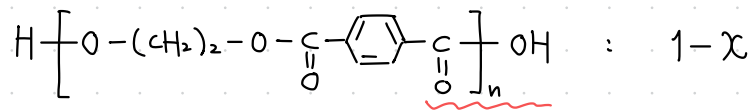
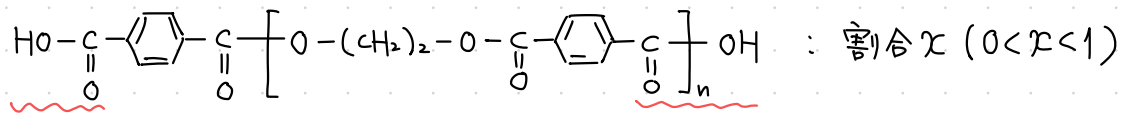
16)3



$$\frac{0.946 \text{ g}}{M \text{ g/mol}} \times \frac{50 \text{ mL}}{100 \text{ mL}} \times \underset{\text{H}^+ \text{ mol}}{1} = 5.0 \times 10^{-3} \text{ mol/L} \times \underset{\text{OH}^- \text{ mol}}{\frac{1.57}{1000} \text{ L} \times 1}$$

$$\therefore M = \underline{6.0 \times 10^4}$$

問4



PET 1分子あたり -COOH は、平均  $2x$

$$2 \times x + 1 \times (1-x) = 1+x$$

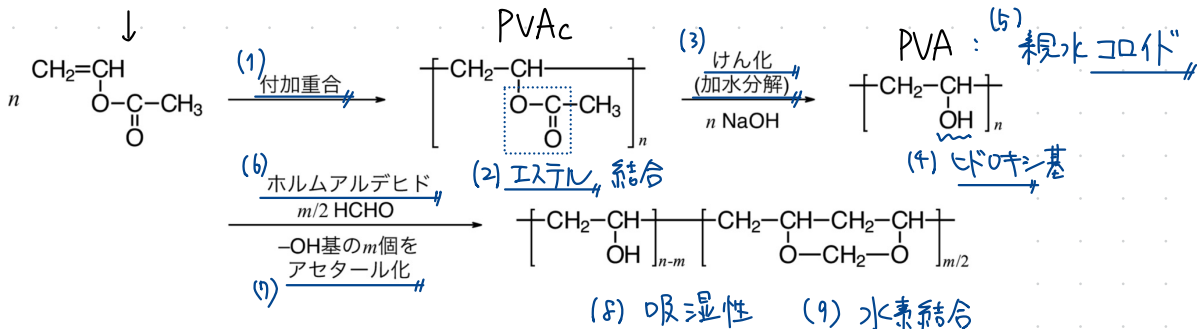
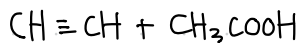
例えば、PET 1g 中の -COOH [mol] は、

$$\frac{1 \text{ g}}{8.0 \times 10^4 \text{ g/mol}} \times (1+x) = \frac{1 \text{ g}}{6.0 \times 10^4 \text{ g/mol}} \quad \therefore x \doteq 0.333$$

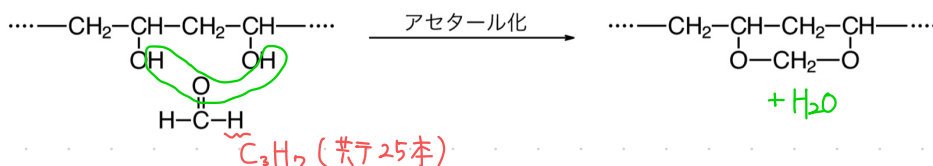
真のPET mol
1分子あたり -COOH 1つのPET mol  
= -COOH mol
33%

問5.  $192n = 8.0 \times 10^4 \quad \therefore n \doteq 417 = \underline{4.2 \times 10^2}$

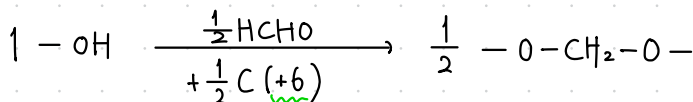
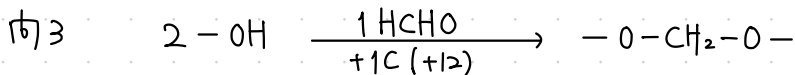
7-4



\* アセタール化



問2 VAはエノール型のため不安定であり、生成しても直ちにアセトアルデヒドに変化してしまうから。



PVAの-OHのうち、割合x (0 < x < 1)だけアセタール化させて生じたビニロンは、

$$\underbrace{44n}_{\text{PVA}} + \underbrace{n \times x \times 6}_{\substack{\text{アセタール化した} \\ \text{-OH数}}} = (44 + 6x)n$$

PVA : ビニロン = 1 : 1 として

$$\frac{500 \text{ g}}{44 \text{ g/mol}} = \frac{w \text{ g}}{(44 + 6 \times 0.4) \text{ g/mol}} \quad \therefore w = \frac{5.27 \times 10^2}{\text{g}}$$

問4 PVAc :  $\text{CH}_2=\text{CH} = 1:1$  式)

$$\frac{172 \text{ g}}{86 \cancel{\text{ g/mol}}} = \frac{92.8 \text{ g}}{(44+6x) \cancel{\text{ g/mol}}} \quad \therefore x = 0.4 \quad \underline{40\%}$$

問5 PVA : HCHO = 1 :  $\underbrace{n \times x \times \frac{1}{2}}_{\text{反応した-OH基}}$  式), HCHO ag  $\approx$  Wg  $\approx$  H $\approx$  2,

$$\frac{172 \text{ g}}{86 \cancel{\text{ g/mol}}} \times \underbrace{\cancel{x} \times 0.4}_{x} \times \frac{1}{2} = \frac{W \text{ g} \times \frac{30}{100}}{30 \text{ g/mol}} \quad \therefore W = \underline{40 \text{ g}}$$

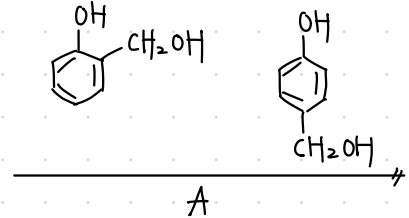
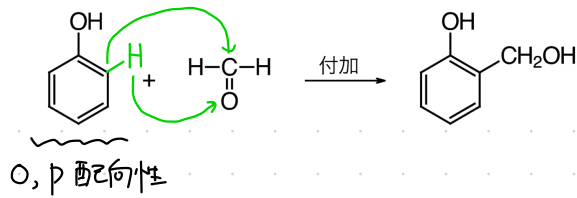
$\text{PVAc mol} = \text{PVA mol}$   $\text{HCHO g}$

7-5

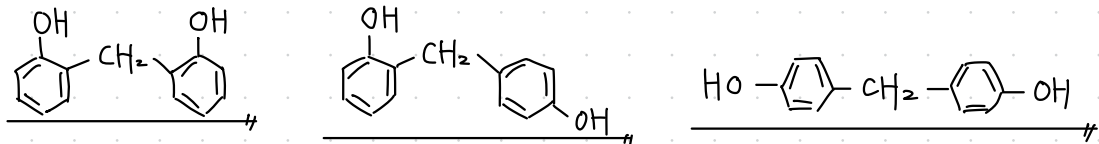
問1 酸 cat : /ボラック

塩基 cat : シソール

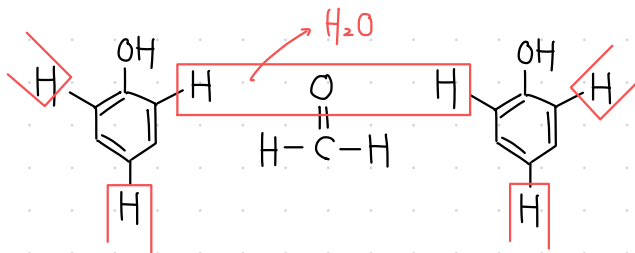
問2 付加反応を考えた。



問3 縮合反応を考えた。



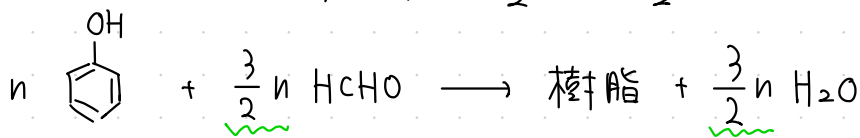
問4.



フェノール中の  $-H \times 2$  と  $HCHO \times 1$  分子が反応

→ フェノール中の  $-H \times 1$  と  $HCHO \times \frac{1}{2}$  分子が反応

→ フェノール 1 分子あたり,  $HCHO \frac{1}{2} \times 3 = \frac{3}{2}$  分子が反応



94 g

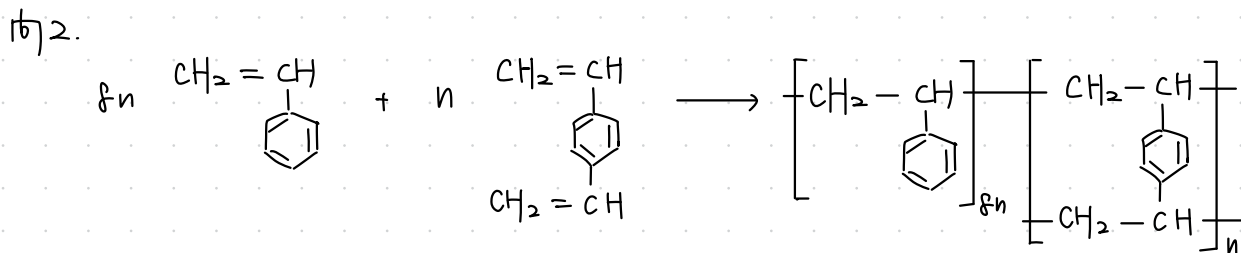
45 g

$$\frac{45g}{30 g/mol} \times 18 g/mol = 27 g$$

$\therefore \text{樹脂} = 94 + 45 - 27 = 112 g$

$7 - \boxed{6}$ 

問1 反応:  $p\text{-}\text{C}_6\text{H}_4\text{NO}_2 + \text{C}_6\text{H}_5\text{NH}_2 \rightarrow \text{p-}\text{C}_6\text{H}_4\text{NHC}_6\text{H}_5 + \text{H}_2\text{O}$   $= \frac{8.32 \text{ g}}{104 \text{ g/mol}} \cdot \frac{1.30 \text{ g}}{130 \text{ g/mol}} = 8:1$



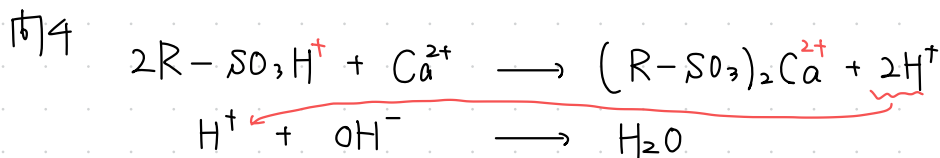
$$104 \cdot 8n + 130n = 8.0 \times 10^4 \quad \therefore n = 83.2$$

$$\therefore p_n = 665.6 = \underline{6.7 \times 10^2}$$

分子量 =  $8.0 \times 10^4$  +  $\frac{8n \times 80}{665}$   $\div 1.3 \times 10^5$   
 スルホン化前  
 スチレン単位

もとの共重合体: (オン交換樹脂 = 1:1 型)

$$\frac{50 \text{ g}}{8.0 \times 10^4 \text{ g/mol}} = \frac{W \text{ g}}{1.33 \times 10^5 \text{ g/mol}} \quad W = 83.75 \text{ g}$$



$$\text{Ca}^{2+} (\text{CaCl}_2) : \text{OH}^- (\text{NaOH}) = 1 : 2 \text{ F1}$$

$$x_{\text{mol/L}} \times \frac{10}{1000} \text{ L} \times 2 = 0.10 \text{ mol/L} \times \frac{40}{100} \text{ L}$$

$$x = \underline{0.20 \text{ mol/L}}$$



問5. 樹脂に含まれる  $-SO_3H$  は,

$$\frac{5.61 \text{ g}}{1.33 \times 10^5 \text{ g/mol}} \times \underbrace{86}_{665} = \underline{2.80 \times 10^{-2} \text{ mol}}$$

樹脂 mol      ↑足す

通じた  $Na^+$  は,

$$0.50 \text{ mol/L} \times \frac{100}{1000} \text{ L} = 5.0 \times 10^{-2} \text{ mol}$$

$-SO_3H \rightleftharpoons Na^+ = 1:1$  より,  $Na^+$  不足あり.

$$[H^+] = \frac{2.80 \times 10^{-2} \text{ mol}}{0.4 \text{ L}} = 7 \times 10^{-2} \text{ mol/L.}$$

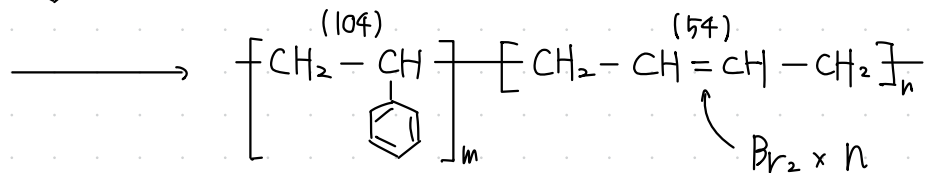
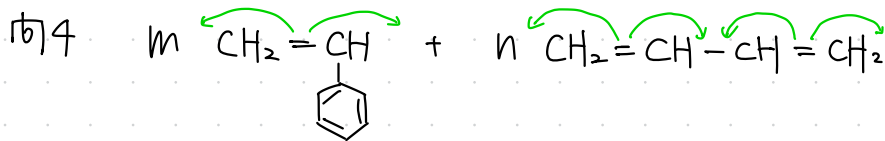
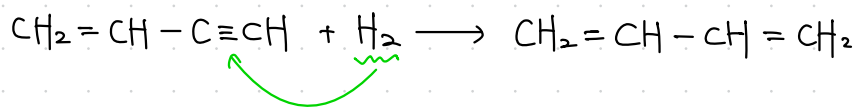
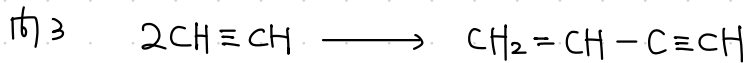
$$pH = 1.15 \div \underline{1.2}$$

7-8

問1 (1) イソプレン (2) 加硫



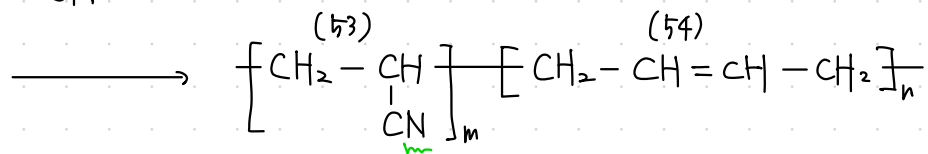
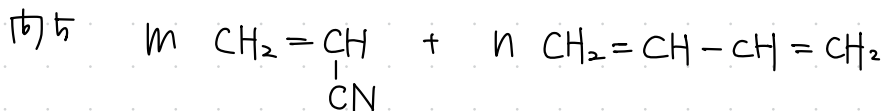
問2 C原子間2重結合が酸素と反応して酸化開裂をおこなうため。



SBR :  $\text{Br}_2 = 1 : n$  式)

$$\frac{1 \text{ g}}{104m + 54n \text{ g/mol}} \times n = \frac{2 \text{ g}}{160 \text{ g/mol}} \quad \therefore 4m = n$$

$$\therefore m : n = 1 : 4$$



条件式)

$$\frac{14m}{53m + 54n} = \frac{8.75}{100} \quad \therefore 2m = n \quad \therefore m : n = 1 : 2$$

ブタジエン : NBR =  $n : 1 = 2m : 1$

$$\frac{w \times 10^3 \text{ g}}{54 \text{ g/mol}} \times \frac{1}{2m} = \frac{10 \times 10^3 \text{ g}}{53m + 54 \cdot 2m \text{ g/mol}}$$

$$\therefore w \div 6.7 \text{ kg}$$